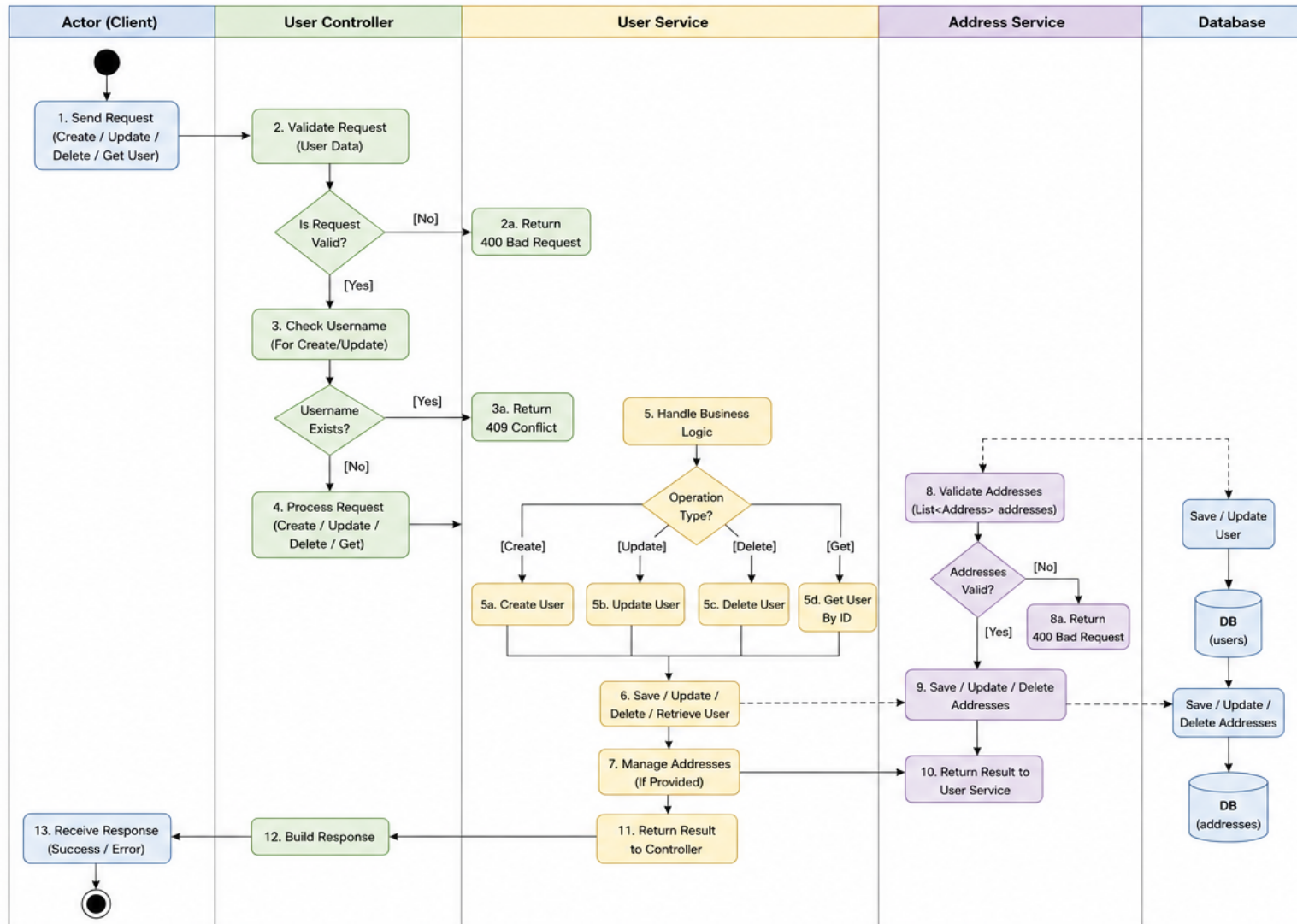


Activity Diagram – Spring Boot Project (User & Address Management) with OCL

Goal: Manage users and their addresses in the system.



Note:
This activity diagram follows OCL constraints, applies guards on decisions, enforces business rules, and defines pre/postconditions for the operations.

OCL Constraints

- context User inv:**
self.username->notEmpty() and self.username->size() >= 3
- context User inv:**
self.email->notEmpty() and self.email->matches('.*@.*\ \\.*)
- context Address inv:**
self.street->notEmpty() and self.city->notEmpty() and self.state->notEmpty() and self.zipCode->notEmpty()
- context User inv:**
self.addresses->forAll(a | a.user = self)
- context Address inv:**
self.user->notNull()
- context User inv:**
User.allInstances()->select(u | u.username = self.username)->size() <= 1

Business Rules

- BR1:** Username must be unique.
- BR2:** Email must be in a valid format.
- BR3:** A user can have one or more addresses.
- BR4:** An address must belong to exactly one user.
- BR5:** Deleting a user will also delete all their addresses.
- BR6:** All fields in Address are mandatory.

Decision Conditions (Guards)

- [Is Request Valid?]**
[Yes]: All required fields are provided and valid
[No]: At least one required field is missing or invalid
- [Username Exists?]**
[Yes]: User with same username already exists
[No]: Username is unique
- [Operation Type?]**
[Create] : Create new user
[Update] : Update existing user
[Delete] : Delete user
[Get] : Retrieve user
- [Addresses Valid?]**
[Yes]: All address fields are valid
[No]: One or more address fields are invalid or missing

Preconditions / Postconditions

Pre:

- Client is authenticated (if required).
- Request payload follows the constraints.

Post:

- User is created/updated/deleted/fetched successfully.
- Addresses are created/updated/deleted accordingly.
- Response is returned to the client.